Minghan Lyu

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EDUCATION

SHANDONG University (SDU), Jinan, China **Sep2023 - Now**

I am an undergraduate student in Computer Science at the School of Software, Shandong University (SDU), and a member of the New Engineering AI Innovation Experimental Class under the Joint SDU-NTU Centre for Artificial Intelligence Research (C-FAIR).

Related courses: Numerical Computing (92), Information Retrieval (92), Linear Algebra (91), Computer Organization and Architecture (90), Introduction to the Artificial Intelligence Major (95), Social Networks and Recommender Systems (90)

RESEARCH EXPERIENCE

Machine Learning for Protein Modeling **Jun2026-Sep2026**

Advisor: Sorin Istrail (Professor)

- Led a three-member course project on CGnet/CGSchNet transferability; developed the CGSchNet analysis and integrated the final report and presentation.
- Independently audited published data for 27 ubiquitin mutations and two protein fluctuation profiles using bootstrap, sensitivity, and structure-localization analyses.
- Built a separate ATLAS benchmark from 200 protein chains with three MD replicas each and an ECOD-component-disjoint 132/38/30 train/validation/test split.
- Combined contact-persistence prediction with weighted Gaussian network models; achieved test persistence MAE 0.0856 and Spearman 0.8878.
- Improved mean test RMSF Spearman from 0.7170 to 0.7229 over uniform GNM; validated the modest gain with component-level resampling and ablations.

Policy-Guided Mango Shape Annealing **Apr2026-Aug2026**

Advisor: John H. Reif (Professor)

- Reframed Mango's stochastic shape-annealing search as masked, constraint-aware proposal guidance for wireframe DNA origami design.
- Engineered a hierarchical grammar-action interface over rule, site, and parameter choices with legal action masking and failure diagnostics.
- Built reproducible B1/B2/B3 benchmark pipelines with unified logging, constraint-failure analysis, candidate audits, and baseline comparisons.
- Demonstrated PG-Mango gains over Mango ShapeAnneal and Random-Feasible on 5/5 B1-family benchmarks while preserving manufacturability constraints.
- I have open-sourced this project on GitHub.

Eukaryotic Gene Structure Prediction with HMM **Oct2025-Jan2026**

Advisor: Jun Wang (Professor)

- Framed gene boundary inference as repeat-sensitive genome sequence labeling.
- Engineered FASTA/GFF3 preprocessing to produce auditable aligned samples and splits.
- Implemented a multi-state HMM with CDS phase modeling and standardized GFF3 decoding.
- Quantified nucleotide/exon Sn-Sp to guide iterations and targeted refinements.
- I have open-sourced this project on GitHub.

Traffic Forecasting on METR-LA **Seq2025-Dec2025**

Advisor: Yongshun Gong (Professor)

- Tackled multi-step traffic forecasting under non-stationary dynamics and missing observations.
- Built an end-to-end pipeline from cleaning and windowing to masked metric evaluation.
- Modeled spatiotemporal dependencies with graph priors and multi-branch sequence learning.
- Reduced test MAE/RMSE/MAPE to 3.57/7.37/10.17% on METR-LA.
- I have open-sourced this project on GitHub.

News Recommendation with NPA **Jan2025-Jun2025**

Advisor: Li Lian (Associate Professor)

- Reproduced NPA as a reliable baseline for personalized news ranking on MIND.
- Built end-to-end impressions and user-history pipelines with controlled negative sampling.
- Enhanced text representations with Transformer-style encoders for deeper semantic matching.
- Improved offline ranking metrics (AUC/MRR/nDCG@K) with stable training and tuning.

LLM-based Paper Screening **Nov2024-Dec2024**
Advisor: Yuqing Sun (Professor)
- Addressed slow and inconsistent systematic-review screening for titles and abstracts.
- Designed structured prompts aligned with inclusion/exclusion criteria and project constraints.
- Applied LLM-based classification with reproducible logging and lightweight quality checks.
- Increased screening throughput while maintaining consistent decision quality.

PROJECT EXPERIENCE

Yantu Lingban: Agentic Exam-Preparation Companion System **Mar2026-Jun2026**
Advisor: Ning Liu (Assistant Researcher)
- Constructed an agentic Windows desktop companion system for postgraduate-exam preparation, covering study planning, subject question answering, mistake review, proactive care, and emotion/diet management.
- Designed a multi-agent workflow with a central orchestrator, specialized study/life/psychology agents, and a shared user-state center for context-aware decisions.
- Integrated rule-driven triggers for active interventions, spaced mistake review, and adaptive plan adjustment based on learning progress, pending review tasks, and user status.
- Connected Electron desktop functions, FastAPI services, WebSocket streaming, OCR-based screenshot question answering, and report generation into end-to-end study loops.
- I have open-sourced this project on Gitee.

Multimodal Understanding for Product Live-stream **Sep2025-Dec2025**
Advisor: Lei Meng (Professor)
- Diagnosed noisy live-stream content and weak cross-modal alignment for product retrieval.
- Built a reproducible pipeline to align keyframes, speech text, and product metadata.
- Mined hard negatives via embedding similarity to sharpen fine-grained contrastive learning.
- Reduced background noise with ROI localization and improved fusion stability via late ensembling.
- I have open-sourced this project on GitHub.

Brand-Geographic Grid Prediction **Mar2025-Jul2025**
Advisor: Meng Chen (Associate Professor)
- Formulated site selection as next-grid prediction from sparse, weakly timestamped trajectories.
- Engineered grid-aligned sequences with density-aware ordering and sliding-window supervision.
- Built a multimodal predictor fusing trajectory encoding with geographic/context features.
- Achieved Acc@1/5/10=0.270/0.567/0.724 and MRR=0.450 on test data.

INTERNSHIP

Invoice Reimbursement Review System **Jul 2026**
Research and Practical Teaching Center, Chinese Aeronautical Establishment
Advisor: Siwei Chen
- Developed browser-based student submission and local Python teacher review workflows for reimbursement materials.
- Integrated PDF/Word parsing, OCR, and optional vision-language models for invoice extraction and supporting-document checks.
- Implemented invoice-level reconciliation, manual review, Excel/PDF exports, and portable Windows packaging.
- Assisted Coze-based AI programming labs for Khalifa University exchange students.

PUBLICATION

Lyu, M. (2026). Predicting Molecular Dynamics Contact Persistence from Static Structures for Weighted Gaussian Network Models. Theoretical and Natural Science, 190, 142-154.
ICBioMed 2026 symposium: Biomedical Engineering and Smart Medicine (ICBM_HZ 2026).
DOI: 10.54254/2753-8818/2026.HZ37271

AWARDS AND HONORS

- 2024 Mathematical Modeling National Contest, Problem C, Provincial First Prize(8%)
- 2025 Mathematical Modeling National Contest, Problem C, Provincial First Prize(8%)
- Innovation Research Specialty Scholarship, 2024-2025 Academic Year
- 2026 COMAP Mathematical Contest in Modeling / Interdisciplinary Contest in Modeling (MCM/ICM), Meritorious Winner (M Award) (6. 67%)

SKILLS AND OTHERS

Programming language: C/C++, Python, Java, MySQL, MATLAB , LaTex ,R
Framework & Tools: PyTorch, Github, TensorFlow, Java IDE, Spring Web, Thymeleaf, Hugging Face, ROS,Ubuntu, Anaconda, BLAST, MEME Suite, Rstudio,Git